

## Code-Layout, Readability and Reusability

|               |                 |
|---------------|-----------------|
| Date          | 29/09/2026      |
| Team ID       | UNM1425EduGenie |
| Project Name  | EduGenie        |
| Maximum Marks | 5 Marks         |

### Code-Layout, Readability and Reusability:

#### Code Layout Checklist

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                                                                                                        |
|------|---------------------------|-----------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | Consistent indentation is maintained in Python, HTML, CSS, and JavaScript code.                                |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | Frontend, backend, and configuration files are organized logically for easy maintenance.                       |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Descriptive variable names are used to make the code easy to understand.                                       |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions and API routes use names related to their learning activities and purpose.                           |
| 5    | Code Comments             | Inline and block comments explain logic       | Partial                       | Comments are used where required to explain important sections and logic.                                      |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Learning activities such as Q&A, explanation, quiz, summarization, and recommendations are handled separately. |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | Unnecessary duplicate code is avoided to keep the codebase clean and maintainable.                             |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           | API and request-related errors are handled and appropriate responses are returned to the user.                 |

#### Reusable Components / Modules

| S.No | Component / Module Name   | Language / Technology      | Where Reused                                               | Reusability Level (High / Medium / Low) |
|------|---------------------------|----------------------------|------------------------------------------------------------|-----------------------------------------|
| 1    | FastAPI Backend           | Python / FastAPI           | All learning activity API requests                         | High                                    |
| 2    | Gemini API Integration    | Python / Google Gemini API | Q&A, explanation, quiz, summarization, and recommendations | High                                    |
| 3    | Learning Activity Modules | Python                     | Different personalized learning functions                  | High                                    |

|   |                               |                         |                                            |        |
|---|-------------------------------|-------------------------|--------------------------------------------|--------|
| 4 | Web Interface Components      | HTML / CSS / JavaScript | Student interaction and result display     | Medium |
| 5 | API Request/Response Handling | FastAPI / JSON          | Communication between frontend and backend | High   |

Overall Code Quality Assessment

| Aspect                   | Rating (1-5) | Comments                                                                                               |
|--------------------------|--------------|--------------------------------------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | The code is logically organized into frontend, backend, and learning-related components.               |
| Readability              | 4            | Clear naming, consistent formatting, and simple structure make the code easy to understand.            |
| Reusability              | 5            | Common backend functions and Gemini API integration can be reused across multiple learning activities. |
| Documentation / Comments | 4            | Important logic is documented with comments where necessary.                                           |
| Overall Score            | 5            | The EduGenie codebase follows a clean, modular, readable, and maintainable structure.                  |